Arjun Narang Final Project Modeling Report

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Objective

The goal of this modeling section is to predict the response variable (Resp) using individual demographic and employment-related characteristics. Preliminary exploratory analysis revealed:

- Nonlinear patterns between Resp and both age and log(wage)
- Bi-modal and right-skewed distribution, violating strict linear model assumptions
- Meaningful interaction between marital status and both age and log(wage)

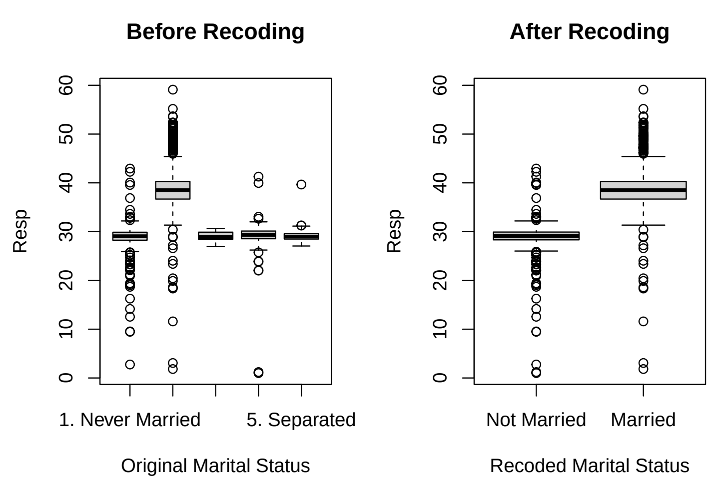
These findings motivated the use of nonlinear regression approaches including polynomial regression, generalized additive models (GAM), and tree-based ensemble methods.

Data Cleaning & Preparation

Removing N/A Values:

We see 60 N/A values for Resp in the 3000 values of data. We will remove them as they were missing completely at random.

Recoding Marital Status:



Marital status values were aggregated into Married vs. Not Married to reflect observed differences in Resp.

Train/Test Split:

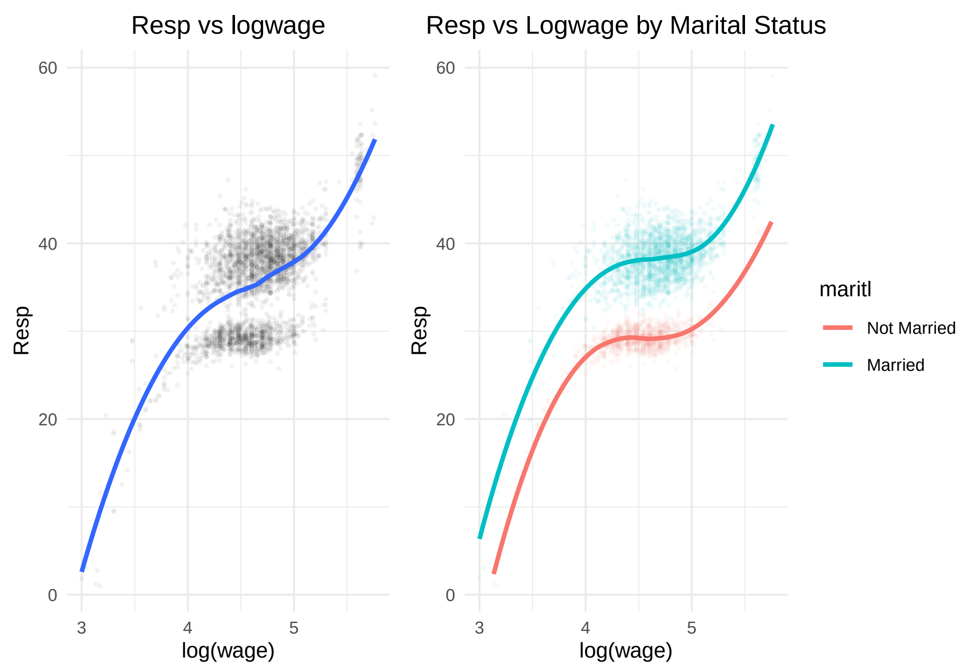
To prepare for the modeling phase, we create an 80/20 train/test split.

```
n <- nrow(data)
train_size <- round(n * 0.8) # Create an 80/20 split
train <- sample(1:n,train_size, replace = FALSE)
test <- -train

Wage.train <- data[train, ]
Wage.test  <- data[test, ]
```

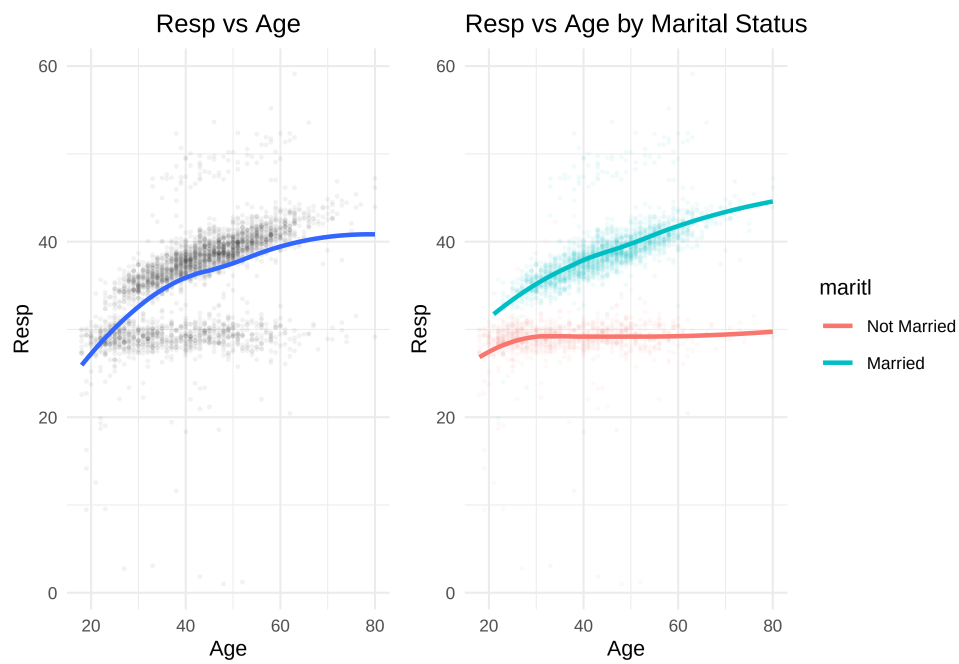
Understanding Interactions

Resp vs log(Wage)



While the model without marital status shows one smooth upward trend, the inclusion of marital status produces two distinct trajectories, especially at higher wage levels, illustrating a meaningful interaction effect.

Resp vs Age



Including marital status reveals an interaction effect, where married individuals display consistently higher and more rapidly increasing response values across age.

Modeling Strategy

To evaluate predictive performance and capture the nonlinear and subgroup-dependent patterns observed in the exploratory analysis, multiple modeling approaches were fit in increasing order of flexibility.

Polynomial Regression:

Polynomial regression was first used as a simple and interpretable way to introduce curvature in the model.

- Captures curvature without fully abandoning linear structure
- Used as a first benchmark for nonlinearity
- Fit with and without marital-status interaction terms
- Helps determine whether interaction effects meaningfully improve model accuracy

Generalized Addative Models (GAM):

GAMs were implemented to address the limitations of polynomial models by learning smooth, data-driven functions rather than relying on fixed polynomial shapes.

- Allows for smooth functions rather than assuming polynomial shape
- Supports different smooths by marital status, aligning with observed interaction effects
- Provides interpretability while modeling nonlinear patterns more accurately
- Iteratively expanded to include additional categorical predictors (education, race, jobclass)

Tree Based Methods:

Tree-based approaches were considered to compare parametric models against nonparametric methods capable of automatically detecting interaction structures.

- CART, bagging, random forests, and boosting were tested
- Automatically capture nonlinearities and interaction structures
- Useful as a performance benchmark against GAM
- Less interpretable; performance gain must justify complexity

Model Evaluation Critera:

Model evaluation focused on predictive performance and stability across validation folds.

- Primary metric: Test Mean Squared Error (MSE) on held-out 20% test sample
- Secondary: 10-fold cross-validation for GAM to assess stability
- Considered interpretability when comparing final candidate models
- Model selection balances accuracy, complexity, and real-world explanation

Polynomial Regression Results

Polynomial regression models were first fit to establish a baseline nonlinear relationship before combining them to search for the lowest test MSE.

Table 1: log(Wage) and Age Polynomial Test MSE Results

Model	Test_MSE
Age Polynomial (Degree 2)	23.124524
log(Wage) Polynomial (Degree 3)	19.075852
Age Polynomial (Degree 2) with Marriage Interaction	8.068780
log(Wage) Polynomial (Degree 3) with Marriage Interaction	4.043070
Age and log(wage) Polynomial (no interaction)	14.669246
Age and log(wage) with marital status interaction	1.035746

The results show that adding nonlinear terms for both predictors does not improve performance by much unless interaction effects are included, indicating that subgroup differences by marital status drive much of the predictive gain.

Introducing Categorical Predictors

In an effort to improve the model's performance, the age and log(Wage) model with interaction effects is set as the baseline. From there various categorical values were tested to gauge an improvement in Test MSE.

Table 2: Top 5 Polynomial Models by Test MSE

Model	Test_MSE
Base + education + race + jobclass + health + health_ins	0.976
Base + education + jobclass + health + health_ins	0.977
Base + education + race + jobclass + health_ins	0.977

Base + education + jobclass + health_ins	0.978
Base + education + race + jobclass + health	0.979

The top-performing polynomial models all include the full or near-full set of categorical predictors, suggesting that demographic and job-related characteristics contribute additional explanatory value beyond nonlinear age and wage effects.